[Experiment, Analysis, and Benchmark] Batch Fidelity Certificates for Semantic COUNT: Scope and Full Policy Cost

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***Abstract*—Putting several rows in one language-model prompt can reduce requests for a semantic filter and COUNT, but checking the batch plan also costs work. We empirically audit an existing current-input fidelity checker by connecting its sampled comparisons to the complete query result and policy bill. The study uses native LOTUS filtering and a declared custom batch operator in 36 configurations over six workloads, two text domains and two models, with 16 additional policies comparing checking budgets. Our joint benefit criterion requires a delivered batch result, at most 5% row disagreement with each of two separate singleton executions, fewer requests and fewer raw input-plus-output tokens. Unchecked random and similarity batches meet it in one and two of the 12 primary configurations, respectively; the checked policy meets it in none. Source COUNT accuracy is a separate outcome. The sole accepted deployment meets observed fidelity tolerance, but calibration turns its token saving into a 3.01% increase, despite 79.69% fewer requests. Raising the budget from 80 to 320 probes leaves every follow-up policy in singleton fallback with a higher bill. All 48 cache-enabled fixed-plan executions use less raw input but more uncached input than their singleton baselines; all 24 cache-disabled executions use less of both. These cache rankings apply to the declared repeated-predicate representation. Distinguishing delivered labels, output format, reference fidelity, source COUNT and complete costs makes the bounded plan decisions reproducible.**

***Index Terms*—semantic query processing, large language models, COUNT, fidelity certification, prefix caching, query optimization**

I. Introduction

A database query asks how many reviews in a critic cohort are clearly positive. After selecting the cohort, the engine applies a language-model semantic filter to each review and computes `COUNT(*)` from the returned Boolean labels. One physical plan sends one row per prompt; another puts several rows in each prompt. Fewer requests can make batching attractive, but the labels can change. A fidelity checker therefore spends additional model calls comparing sampled batch labels with one-row, or *singleton*, labels before selecting a batch size.

The optimizer must decide whether the checked plan is worth executing. That decision requires connecting three things: what the check certifies, what the complete deployed plan actually returns, and what the whole policy costs. A check uses small artificial probe batches; deployment partitions the full input. Agreement with singleton labels measures fidelity to an execution reference, whereas agreement with supplied source labels measures task accuracy. The bill includes checking and any singleton fallback. With prefix caching, the number of input tokens written in prompts also differs from the number that require fresh model computation.

These mechanisms have established precedents. Company studies context interference, aggregate masking of row changes and cached singleton prefixes [4]; Hassanzadeh and Subramanian measure semantic-query replay variation and cascade calibration [8]. LOTUS defines optimization relative to reference algorithms [5]. NousDB provides the current-input fidelity check studied here and already measures calibration costs, selected sizes and efficient cached singleton execution [6]. Our contribution is an empirical audit that joins the check's observations to a complete deployed result and native cost receipts in one declared engine setting. It measures how those established considerations jointly affect the plan decision.

We ask three questions:

1) **What is preserved?** How do delivered Booleans, strict output validity, fidelity to separate singleton executions, and COUNT accuracy against source labels differ?
2) **Does checking preserve the saving?** Do delivered batch plans stay within 5% row disagreement with each

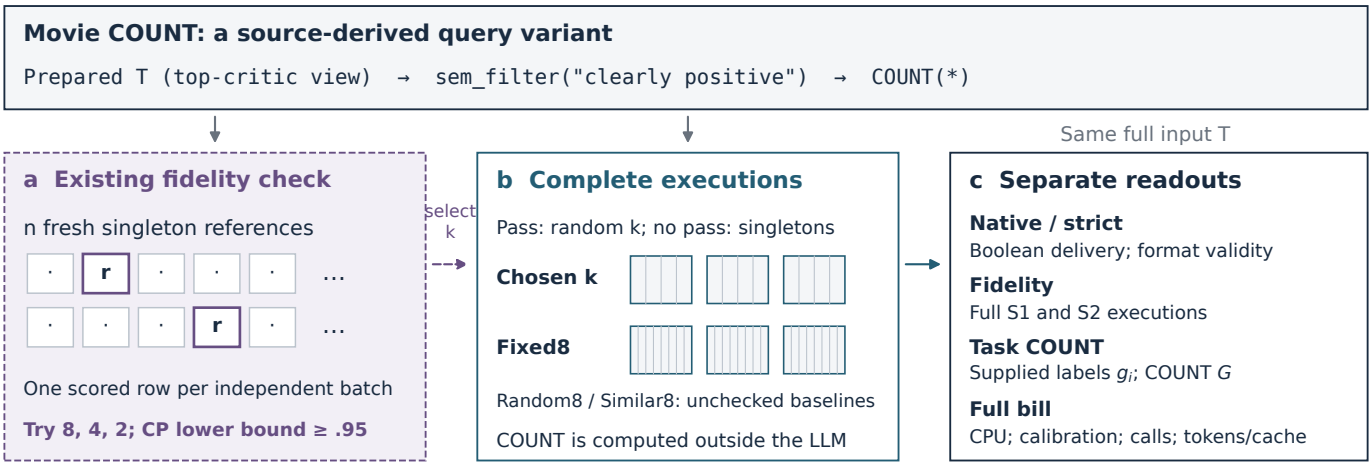


Fig. 1. From a semantic COUNT query to a checked execution. The prepared review relation is the input to both checking and deployment. (a) The current-input check uses a Clopper–Pearson (CP) bound and compares one focal row per independently sampled probe batch with a singleton label. Its nominal certificate concerns agreement under that sampler. (b) Deployment labels every row once in disjoint batches, then computes COUNT outside the model. (c) We separately measure Boolean delivery and strict format, agreement with two complete singleton executions (S1/S2), COUNT error relative to the supplied-source count ($G = \sum_i g_i$), and the complete policy bill. The drawing is schematic; custom batching changes the prompt and output contract.

reference while reducing complete requests and raw input-plus-output tokens? What changes with a larger checking budget?

3) **Which cost measure orders the plans?** Do raw input and native uncached-input counters favor the same plan when prefix caching is enabled?

We execute native LOTUS filtering and deterministic COUNT with a custom multirow operator. The 36-configuration matrix spans six workloads, two domains and 512–7,600 rows using Qwen3-4B-Instruct-2507 (Qwen) and GLM-4-9B-Chat (GLM). Twelve primary configurations compare every workload at a common size/cache setting; 24 anchors vary size or cache. Sixteen follow-up policies compare 80/320-probe budgets. Complete singleton replays, deployed partitions and native receipts connect checking to results and costs.

Unchecked random and similarity batches meet the joint criterion in 1/12 and 2/12 primary configurations; the checked policy meets it in 0/12. Its sole accepted deployment preserves both observed references, but calibration consumes its raw-token saving. Larger follow-up budgets retain fallback at higher cost. Cache-enabled fixed plans favor batching on raw input and singleton execution on uncached input. Classical COUNT controls and earlier experiments place these representation-specific observations in context. The artifact preserves every configuration, negative case and resource record.[1]

II. What the Query and Check Preserve

A. Delivered Labels and Observed References

Let $T = (r_1, \ldots, r_N)$ be the prepared input relation. Each row occurrence has a distinct ID and its complete source-provided text field. A deployed plan labels every occurrence exactly once. A probe batch is an artificial context used for checking; probe batches can overlap and need not cover T (Fig. 1).

Native delivery means that the engine returns a complete Boolean vector and hence a COUNT. LOTUS's native parser searches for `true` before `false` after the first `Answer:`, or in the whole response if that prefix is absent; it defaults to `True` if neither occurs. We also apply an *extra strict-format check*: only a standalone Boolean, optionally prefixed by `Answer:`, is valid. Its label is $s_i \in \{0, 1, \perp\}$, where $\perp$ denotes invalid format. Thus a query can deliver a native COUNT while its strict COUNT is unavailable. We report format validity, actual defaults and execution exceptions separately.

Preserved Qwen World S1 responses include `Answer: Sports` and `Answer: Sci/Tech`. Neither contains a Boolean word, so native parsing defaults to `True` while strict validity fails. The custom multirow operator instead requests a constrained JSON object mapping ordered local row IDs to `True`/`False` strings. All k required labels must be valid; a missing or invalid label makes the complete vector and COUNT unavailable.

S1 and S2 are two complete native singleton executions with fresh responses, used only to evaluate completed policies. Comparing with both exposes sensitivity to the observed reference execution. The checker pays for separate calibration singleton calls; neither S1/S2 nor source labels enter selection. Held-out validation also obtains fresh singleton/batch pairs. These are observed execution references, not ground truth. For a complete policy vector b, its native-reference row losses are

$$d_j(b) = \frac{1}{N} \sum_i \mathbf{1}[b_i \neq n_i^{(j)}], \quad j = 1, 2, \tag{1}$$

where $n_i^{(j)}$ is reference j's native label. We summarize the

[1]Public materials: https://github.com/yzd2002/semantic-aggregation-audit-icde2027.

larger loss as $d_{\max} = \max(d_1, d_2)$. Strict-vector fidelity is a separate diagnostic, available only when both full strict vectors exist; a valid subset cannot supply a full-relation COUNT.

B. Source COUNT and Row Fidelity

Supplied source labels give a separate task target. Let g_i be the fixed binary source label, $G = \sum_i g_i$ the source COUNT and $C_b = \sum_i b_i$ the plan's COUNT. We report source row error and two COUNT-error units:

$$e_{\text{row}} = \frac{\sum_i \mathbf{1}[b_i \neq g_i]}{N}, \qquad e_N = \frac{|C_b - G|}{N}, \quad e_G = \frac{|C_b - G|}{G} \ (G > 0). \tag{2}$$

$G = 0$ is reported separately. The additive error e_N measures error as a fraction of the input; e_G measures it relative to the source-positive count. Source annotations have not been newly human-verified.

False positives (FP) raise the count and false negatives (FN) lower it. They add in row error but can cancel in COUNT error: the respective numerators are $\text{FP} + \text{FN}$ and $|\text{FP} - \text{FN}|$. Similarly, $|C_b - C_{S_j}|/N \leq d_j$: a realized row-fidelity bound also bounds additive COUNT change to that same realized reference. It does not establish source accuracy or a 5% source-relative COUNT guarantee. Our COUNT controls use the additive tolerance $.05N$.

The check targets a different object: the probability $\varphi(k)$ that a focal batch label agrees with a singleton label under a specified random row/context sampler [6]. A nominal bound on this probability relies on the check's oracle and independence assumptions. It is distinct from d_j for one complete deployed partition; the calibration references and later S1/S2 can also vary. We therefore report calibration, independent validation and deployment separately. Changing to similarity grouping changes the sampler again. A realized above-tolerance loss would require interpretation under these assumptions; it would not by itself establish a false certificate or a NousDB bug.

III. Policies and the Existing Fidelity Check

A. Policy Key

The fixed-plan baselines are **Random8**, a seeded random-permutation partition into batches of eight, and **Similar8**, deterministic term-frequency/inverse-document-frequency (TF–IDF) sparse-cosine greedy grouping with frozen ties. Neither performs calibration or supplies a fidelity certificate. Similar8 is a known grouping reference, rather than the full BATCHER method [1]; its CPU construction is charged.

The checked policies reproduce NousDB's paper-defined Algorithm 2 with a declared tested ceiling $k_{\max} = 8$ supplied to the procedure [6]. We evaluate this subprocedure, rather than NousDB's complete engine or histogram optimizer. Preflight retains complete fields: the longest tested eight-row prompts use 2,260 Qwen and 2,173 GLM tokens. Both plus 128 output tokens fit the 8,192-token window; a length guard also checks every submitted request. Eight is the tested ceiling, not a maximal or optimal size. Candidates are $8, 4, 2$ in descending order, with mismatch tolerance $\eta = .05$ and error budget $\delta = .05$. Names encode the probe budget and agreement rule:

- **Current80-S**, used in all 36 main configurations, permits 80 probes per candidate. Agreement requires valid strict batch and singleton labels that are equal.
- **Current80-N** and **Current320-N**, used in the follow-up, permit 80 and 320 probes. Both require a valid batch label equal to the native singleton Boolean, including disclosed defaults. Strict reference format remains a separate diagnostic.

The S/N suffix distinguishes the strict/native agreement rule. Only the Current80-N/Current320-N contrast holds that rule fixed while changing budget. A post-hoc native-rule replay rescores only original recorded responses. It recovers selection and issued-call bills because native mismatches alone already force every observed rejection (Section VI-D). It runs no new deployment and supplies no quality or latency measurement for an unexecuted policy.

B. Check, Select and Deploy

For a fixed input T and budget n, the procedure is:

1) Draw n focal row occurrences independently and uniformly from T, with replacement, and issue n fresh singleton requests, one per probe index. Even repeated occurrences receive new calls. Reuse this indexed focal/reference sequence across this policy's candidate sizes only.
2) For each candidate $k = 8, 4, 2$, construct each probe batch independently: sample $k - 1$ other occurrences uniformly without replacement, excluding the focal occurrence, and insert it in a uniformly chosen slot. Require the entire k-label JSON response to be valid, then score only the focal label against its singleton reference under the chosen S/N rule.
3) Issue batch probes in chunks of 16 requests. Select the first candidate whose fixed-budget agreement bound passes, accepting only after all n trials. After each whole chunk returns, score every issued trial and reject if even perfect remaining trials cannot pass; then try the next size. Every issued response is scored and billed, including the tail after a logically decisive probe.
4) Physically evaluate the frozen complete random partition at the selected size, obtaining new model responses. If no candidate passes, run fresh native singleton fallback ($k = 1$). Deployment covers every occurrence once and reuses no probe responses.

At $k = 8$, the row grouping is exactly Random8's frozen grouping, evaluated through separate physical calls without reusing Random8 responses. Lower candidates use their frozen partitions. New calls preserve that layout; their statistical independence remains an assumption. Similar8's grouping is different and inherits no certificate from random probes.

Write V_B for validity of the entire k-label batch response and V_S for strict validity of its singleton reference. One trial's

success indicator Z_k requires V_B, V_S and equal focal labels under S; under N it requires V_B and equality to the native singleton label. Then $\varphi(k) = \Pr(Z_k = 1)$ under the stated focal/context sampler and oracle assumptions. This is one focal-trial probability, not a probability that every batch label is correct.

For X_k agreements out of n, the one-sided Clopper–Pearson (CP) lower endpoint is

$$\underline{\varphi}_k = \mathrm{Beta}^{-1}(\delta/3; X_k, n - X_k + 1), \tag{3}$$

with endpoint zero when $X_k = 0$ [24]. The threshold is $\underline{\varphi}_k \geq 1 - \eta = .95$. The $\delta/3$ allocation covers the three candidates in this one policy. Its nominal interpretation retains fixed-oracle and independent-trial assumptions; replay observations do not establish them. It supplies no joint guarantee across the matrix, probe budgets or a post-hoc selected winner.

C. Why Compare Probe Budgets?

NousDB's zero-error feasibility rule [6] gives

$$n_{\min} = \left\lceil \frac{\log(3/\delta)}{\log(1/(1-\eta))} \right\rceil = 80. \tag{4}$$

At this budget, passing requires zero disagreements; 320 probes allow seven. Under a hypothetical independent and identically distributed (IID) mismatch probability q, acceptance probability is $\pi_n(q) = \Pr\{\mathrm{Binomial}(n, q) \leq e_{\max}(n)\}$, where $e_{\max}$ is the allowed disagreement count. At $q = .01$, it rises from .447523 to .983697. This exact binomial calculation explains the potential power gain; it is not measured model behavior. A stopped probe prefix cannot provide an unbiased estimate of q.

Greater power also costs more checking. Calibration uses at most $4n$ requests (singleton references and three candidate families), and deployment at most N, so $C_n \leq 4n+N$. A first-candidate $k = 8$ pass uses $2n+N/8$ requests. At $N = 7,600$, the 80/320 budgets therefore use 1,110/1,590 requests for that pass, or at most 7,920/8,880 with fallback. Each probe batch still reads and labels all k rows. Whether more probes improve the complete bill is an empirical question.

IV. Experimental Design and Cost Accounting

A. Queries, Inputs and Configurations

A *workload* pairs a semantic predicate with its prepared input. A *configuration* fixes that workload, input size, model and cache setting. The primary matrix is six workloads × two models = 12 configurations. Each secondary family is three anchors × two sizes × two models = 12 more (36 total). The separate follow-up is two workloads × two models × two budgets × two repeats = 16 policies, or eight budget pairs (Table I).

Movie has two disjoint critic views, defined by `isTopCritic=True/False`. Both count clearly positive reviews from the supplied `reviewText`, with metadata and sentiment labels withheld from inference. We use version 4 of the official SemBench generator's upstream source [7], [26], deduplicate normalized text globally before forming the views, and fix order by a feature hash. An additional integration case runs original SemBench Movie Q3 unchanged on its official 2,000-row demonstration relation; 120 rows reach the semantic filter. The larger Movie workloads are prepared-relation COUNT variants, not scaled executions of original Q3 on the raw Reviews table.

Throughout, N counts rows entering the semantic filter. The unchanged Q3 integration case is outside both the 36 main configurations and the 16 follow-up policies.

AG News uses a pinned, duplicate-free, feature-hash-ordered 7,600-row test split [25]. Four predicates—World, Sports, Business and Sci/Tech—share this corpus. Each frozen predicate asks for a four-topic classification and then membership in its named topic; Movie uses one clearly-positive question. We retain the complete supplied `text`/`reviewText` fields. They are not claimed to be complete original articles. Acquisition hashes and source notices are retained; AG's license remains unknown.

For example, the frozen World predicate is:

> Classify the news item by its primary topic among World, Sports, Business, and Sci/Tech. Determine if its primary topic is World (international affairs, politics, or world events). News item: {text}

Native LOTUS also specifies: `Answer: <Your answer here. The answer should be either True or False>`. The conflicting topic-word responses are observed outcomes, not a matched causal test of this wording.

Thus six workloads span two domains and three corpus views. Shared-corpus predicates, nested sizes, cache settings and model variants are dependent observations, not independent datasets. Source membership, fields, prompts, groupings and schedules were hash-bound before their corresponding inference; no text was shortened or replaced after outcomes.

B. Engine, Batching and Replays

LOTUS v1.2.4 at a pinned commit executes live `sem_filter` against native vLLM [16] through an isolated CPU process. Native execution builds one prompt per row and submits prompts concurrently. This server concurrency differs from multirow prompting. Our custom batch operator serializes each record's complete native `context_and_claim`, including its repeated predicate, as a JSON string with escaped newlines and quotes. It also changes system instructions and output requirements. The comparison therefore concerns these integrated physical plans; it does not isolate neighboring-text effects or test a minimum-token template that factors out the predicate.

The schematic prompt structures are: native instructions plus one complete context/claim, returning a Boolean; versus batch instructions plus k serialized complete contexts/claims, including k predicate copies, returning local-ID Boolean labels. These describe the representation, not new prompt variants.

Every configuration executes separate S1/S2 references, Random8, Similar8 and Current80-S. Complete plans use no training fillers and cover each input occurrence once. After a main-study non-singleton selection, 80 independently

TABLE I

STUDY DESIGN. EVERY LISTED WORKLOAD/SIZE/CACHE COMBINATION USES BOTH QWEN3-4B-INSTRUCT-2507 AND GLM-4-9B-CHAT. N COUNTS ROWS ENTERING THE SEMANTIC FILTER AFTER STRUCTURED FILTERING. THE FOLLOW-UP CROSSES TWO WORKLOADS, TWO MODELS, TWO BUDGETS AND TWO PHYSICAL REPEATS.

Family	Workloads	Input N	Prefix cache	Count
Primary	Movie top/other critics; AG World, Sports, Business, Sci/Tech	2,048	ON	12 cases
Scale anchors	Both Movie views; AG Sci/Tech	512; 7,600	ON	12 cases
Cache anchors	Same three anchors	2,048; 7,600	OFF	12 cases
Budget follow-up	Movie other critics; AG Sci/Tech	7,600	ON	16 policies

seeded validation probes receive fresh singleton and batch calls at the selected size. Validation never enters selection and is charged only to the research ledger. Fixed-plan results span all 12/36 configurations. A separate same-size grouping comparison with Similar8 is available only for a checked $k = 8$ deployment; other selections change both size and grouping. The follow-up adds no validation family.

C. Follow-Up and Runtime Settings

The follow-up fixes AG Sci/Tech and a feature-hash-selected Movie other-critics view, $N = 7,600$, cache ON, two models, two budgets and two repeats: 16 complete policies. The trigger was the first Qwen AG Sci/Tech result at $N = 7,600$: Current80-S fell back despite strict-valid calibration references. Sci/Tech was an existing anchor; the Movie view was chosen by the minimum seeded hash of the two existing workload IDs, yielding other critics. A fourfold budget tests the zero-versus-seven-allowed-disagreement contrast. The follow-up was frozen before its own calls. Common feature/probe seeds pair the inputs, while all responses and deployments are physically new. Existing S1/S2 are evaluation-only. The repeats are descriptive observations on fixed inputs, not independent datasets.

Each model uses one NVIDIA GeForce RTX 5090 (32 GB; 32,607 MiB reported), PyTorch 2.8.0+cu128, vLLM 0.10.2 and Transformers 4.57.1. Qwen3-4B-Instruct-2507 and GLM-4-9B-Chat [9], [11] run native bfloat16 (BF16) inference with temperature zero, an 8,192-token window, eager execution, at most 32 active sequences and native/batch output caps 32/128. Effective classes, tokenizer/runtime hashes and actual cache-reset receipts are retained. Raw returns are stored before parsing. There is no automatic retry, model replacement, post-label resampling or selective deletion; failures and unknown costs remain explicit.

D. Complete Policy Bills and Reporting Rules

A standalone policy bill includes CPU grouping, fresh calibration references, every issued probe chunk, deployment and fallback. Separate S1/S2 and independent validation are evaluation costs in the research ledger. Each policy starts with an empty prefix cache and can reuse only its own calibration-to-execution cache. Cache OFF is an actual separate runtime configuration. Operator and response memoization are disabled.

Prefix caching reuses computation for a previously processed prompt prefix. It can therefore make repeated singleton prompts cheap even when their logical input-token total is large. For API request a, native counters report input tokens P_a, output tokens O_a and cached input tokens K_a. Let $P = \sum_a P_a$, $O = \sum_a O_a$ and $K = \sum_a K_a$. We distinguish

$$\text{raw tokens} = P + O, \qquad U = P - K. \tag{5}$$

U is *API uncached input*: an observed proxy for fresh input processing (prefill), not exact FLOPs or dollars. Each counter has its own S1 ratio. Output generation is separate; automatic prefix caching reduces prefill rather than decoding [27]. Missing K_a remains unknown with bounds and is never set to zero.

Elapsed time starts after a verified cache reset and includes client isolation, durable tracing, live transport, CPU grouping, calibration and execution. Input preparation, model loading and the reset itself are outside the timer. Shared-server elapsed times are descriptive and support no inferential speed ranking.

For each primary configuration we report delivery, strict validity/defaults, selected size, native and strict S1/S2 losses, source row/COUNT errors and complete costs. The *joint benefit criterion* is descriptive: complete native delivery, $k > 1$, both $d_1, d_2 \leq .05$, fewer requests and fewer raw tokens than S1. Source COUNT, uncached input and elapsed time remain separate outcomes. The strict-vector sensitivity evaluates the same deployment and bill using complete strict plan/reference vectors; it does not rerun selection. Cases are satisfied, evaluated-negative or unavailable, all retained in the 12-case denominator. Secondary and follow-up executions retain their full readouts.

V. CLASSICAL CONTROLS FOR A COUNT-ONLY GOAL

A COUNT-only application can tolerate row errors that cancel, or estimate the count without first labeling every row. We include established finite-population controls with three reported outputs: accept/reject the unchanged COUNT; accept/reject row accuracy; or estimate COUNT directly with an interval and midpoint. These offline controls use supplied source labels, or a recorded singleton vector where available, as a fixed target y. Label lookups count as oracle queries; we do not price them as online model or human judgments.

The main source-label analysis requires a complete *plan strict vector*: all 72 fixed-plan vectors and 26 Current80-S vectors qualify, yielding 98 pairs over the 36 configurations.

Source labels are already complete; this eligibility does not require a strict singleton reference. Strict-reference fidelity in Section II, by contrast, requires both strict plan and reference vectors. A post-hoc source-label sensitivity uses all 108 native plan vectors with the same sample order and fixed look. These eligibility counts do not change the 36-configuration study denominator. Dependence during generation does not invalidate design-based sampling of the fixed vector pair [23]; the guarantee concerns that pair, not another replay or correctness beyond supplied annotations.

A. Auditing a Complete Plan

For a complete realized binary plan vector b, take one simple random sample without replacement of $m = \min(1,024, N)$ target labels. All procedures use this one fixed look, with tolerance $\epsilon = .05$ and error budget $\alpha = .05$. For context, the standard difference estimator $C_b - N\overline{(b-y)}_{\text{sample}}$ adjusts COUNT using sampled errors, following SampleClean/control-variate approximate query processing (AQP) [17], [19]. It is background lineage, not a corrected-count result in our tables. Our *raw-COUNT gate* asks whether uncorrected C_b is already within ϵN of the target.

To construct that gate, divide the plan into predicted-positive and predicted-negative strata. First take the overall random sample, then condition on its observed stratum sample sizes. In stratum h, N_h is population size, F_h total errors, m_h sampled rows and X_h sampled errors. The hypergeometric (HG) law is $X_h \sim \text{HG}(N_h, F_h, m_h)$. Errors are false positives in the positive stratum and false negatives in the negative stratum. Inversion gives population *error-count* intervals $[L_+, U_+]$ and $[L_-, U_-]$, in rows. Allocate α/H to each of H nonempty strata, split across two tails. Empty strata contribute zero; an unsampled nonempty stratum retains $[0, N_h]$. A sole nonempty stratum uses full α.

On the same joint event, intervals for signed COUNT error and total row error are

$$I_\Delta = [L_+ - U_-, U_+ - L_-], \quad I_{\text{row}} = [L_+ + L_-, U_+ + U_-]. \tag{6}$$

The raw-COUNT gate passes if $I_\Delta \subseteq [-\epsilon N, \epsilon N]$; the *joint-strata row gate* passes if $\sup I_{\text{row}} \leq \epsilon N$. A separate direct row baseline inverts an unstratified two-sided HG interval for sample mismatches; neither row method is claimed uniformly tighter.

B. Estimating COUNT Directly

Direct AQP samples the target labels without requiring a complete batch vector. It independently inverts the sample's target-positive count to obtain a two-sided HG interval, reports the midpoint, and checks interval width $\leq 2\epsilon N$. It costs m oracle queries and no full batch inference, so remains defined even if a batch plan cannot return a complete vector.

Each method/vector pair has a separate nominal 95% scope. The single look is not repeatedly inspected until passing [22], [23], and selecting a favorable method or plan afterward does not preserve joint 95% coverage. An S1-based decision evaluated against S2 is a replay sensitivity, not a coverage failure against S1. Both COUNT-error normalizations are reported, while the gates use additive $.05N$ throughout.

VI. Live Execution Results

All 36 main configurations and 16 follow-up policies completed. Figure 2 summarizes the primary matrix. The main checker selected a batch size above one only for GLM Sports at $N = 2,048$, cache ON. We follow that execution from checking to the returned count and full bill before examining the remaining outcomes.

A. Walkthrough: A Passed Check and Its Full Bill

For GLM Sports, Current80-S selected $k = 8$ after all 80 calibration comparisons agreed. The separately seeded validation then found two mismatches in 80 fresh comparisons (.025). Deployment labeled all 2,048 rows in 256 disjoint batches and disagreed with each complete S1/S2 reference on 17 rows (.00830). These probe and partition fractions have different denominators; all meet the .05 observed tolerance here. The accepted execution supplies no observed certificate-transport failure.

The returned count was 519 against source $G = 517$, giving $e_N = .10\%$ and $e_G = .39\%$, with 1.66% source row error. S1 itself had 2,025/2,048 correct source labels. This is therefore a usable-reference case for studying costs, independently of the Qwen failure panels below. The separately executed Similar8 plan had 13/2,048 reference mismatches (.00635); its changed grouping is an observed alternative, not an additional certificate from random probes.

Deployment alone used 256 requests and 255,425 raw input-plus-output tokens. Calibration added 160 requests and 93,120 tokens. The complete policy thus used 416 requests and 348,545 tokens, compared with S1's 2,048 requests and 338,365 tokens. Requests fell by 79.69%, but raw tokens rose by 3.01%. The separately executed Random8 baseline had the same observed deployment-only token bill, .754880 times S1. Charging the check consumes that saving, so the checked plan fails the joint benefit criterion despite meeting observed fidelity tolerance.

B. Reference Fidelity and Source COUNT Are Different

Table II reports S1's source quality for every primary workload. Qwen World and Sports triggered default-True parsing on 1,936 and 1,664 of 2,048 rows. Their native counts were 1,999 versus source $G = 514$, and 1,741 versus $G = 517$; source row errors were 74.37% and 60.74%. A post-hoc World S1 inspection found normal `stop` termination on all 2,048 calls with only 4–6 output tokens. These observations characterize the frozen two-step query wording, model and native parser together; no matched rewrite isolates the cause. Strict-format checking exposes the format issue while native COUNT remains delivered. Preserving such a reference would also preserve its task errors.

All 12 primary Current80-S policies delivered native counts, but only six had complete strict vectors (Table III). The native joint benefit result is 0/12. The strict-vector sensitivity has

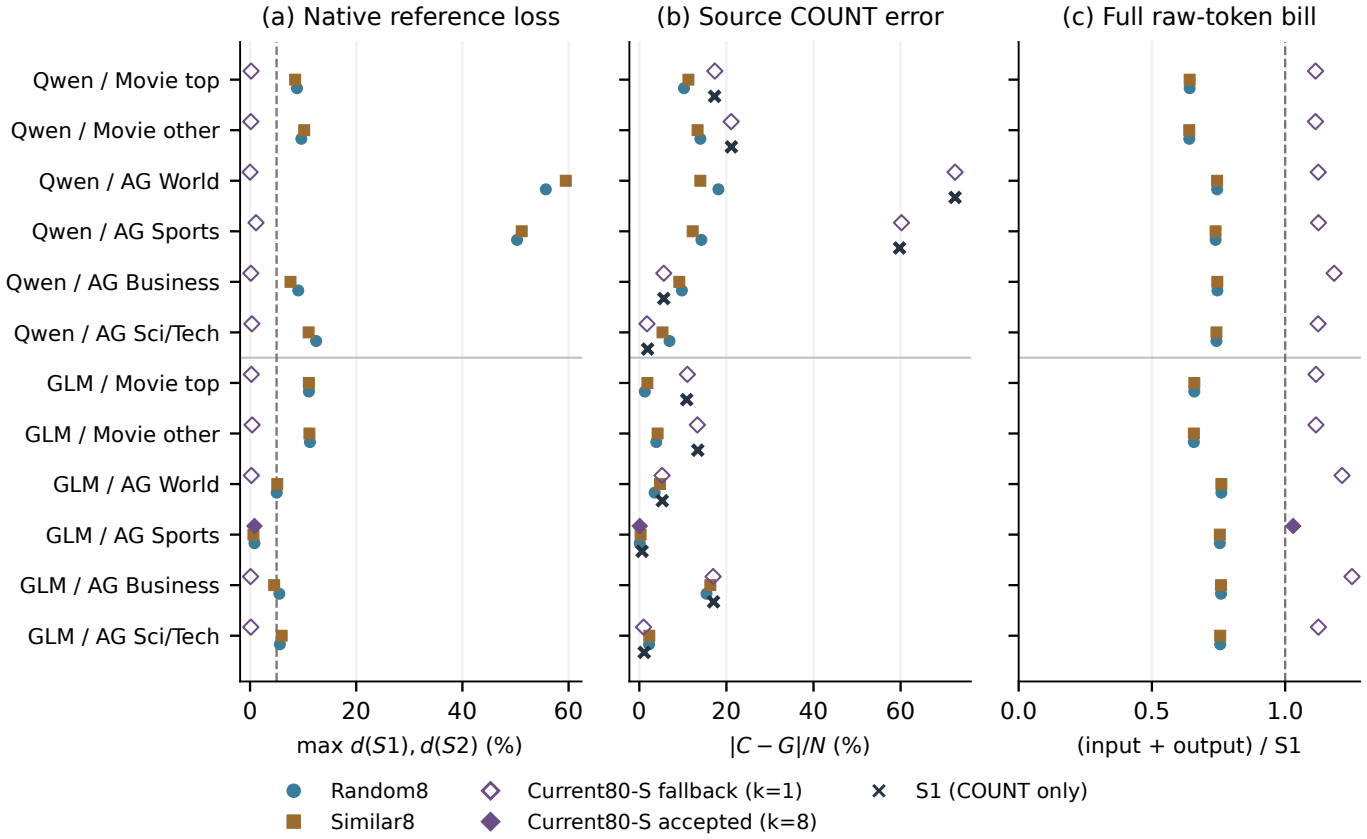


Fig. 2. All 12 primary configurations ($N = 2,048$, cache ON). Hollow purple diamonds are Current80-S singleton fallbacks ($k = 1$); the filled diamond is the sole accepted batch deployment ($k = 8$). Panels show (a) maximum row disagreement with S1/S2, (b) source COUNT error divided by N, and (c) complete calibration-plus-execution raw-token bills relative to S1. Dashed lines mark the 5% reference tolerance and unit S1 raw-token cost. Low fallback loss is not evidence of a certified batch. Fixed batches use 256 requests, one eighth of S1.

TABLE II
NATIVE SINGLETON REFERENCE S1 AGAINST SUPPLIED SOURCE LABELS IN ALL PRIMARY CONFIGURATIONS ($N = 2,048$, CACHE ON). G: SOURCE COUNT; C: DELIVERED NATIVE COUNT; D: DEFAULT-TRUE PARSES; R: SOURCE ROW-ERROR PERCENTAGE. NATIVE DELIVERY AND REFERENCE FIDELITY DO NOT IMPLY TASK CORRECTNESS.

Workload	G	Qwen C	Qwen D/N	Qwen R (%)	GLM C	GLM D/N	GLM R (%)
Movie top	1,251	897	0/2048	19.92	1,028	0/2048	16.36
Movie other	1,418	985	0/2048	22.61	1,143	0/2048	17.04
AG World	514	1,999	1936/2048	74.37	622	4/2048	8.59
AG Sports	517	1,741	1664/2048	60.74	530	0/2048	1.12
AG Business	498	613	7/2048	10.69	847	2/2048	18.99
AG Sci/Tech	519	480	21/2048	9.13	496	0/2048	8.25

zero satisfied, six evaluated-negative and six unavailable cases. Unavailable strict COUNT is distinct from native query failure.

The unchanged SemBench Q3 integration case illustrates the denominator distinction. Its 2,000-row demonstration relation sends $N = 120$ rows to the semantic filter, with $G = 14$. Qwen returns 7 and GLM 9 under both cache settings, without defaults and with complete strict outputs. The absolute COUNT errors are seven and five rows: $e_N = 5.83\%$ and 4.17%, but $e_G = 50.00\%$ and 35.71%. These integration measurements are separate from the larger prepared-relation variants.

C. Whole-Policy Benefit Across the Primary Matrix

Figure 2 separates native-reference loss, source COUNT error and full raw-token cost for all 12 primary configurations. Table IV applies the fixed-denominator joint benefit criterion from Section IV-D. Random8 meets both reference-loss thresholds and lowers requests and raw tokens in GLM Sports (1/12). Similar8 does so in GLM Sports and Business (2/12); Business's maximum reference loss is 4.54%. These fixed plans have neither calibration cost nor a fidelity certificate.

Current80-S meets the criterion in 0/12 primary configurations, including 0/10 outside the two Qwen failure panels.

TABLE III

CURRENT80-S IN ALL 12 PRIMARY CONFIGURATIONS ($N = 2,048$, CACHE ON). V_s: STRICT-VALID PLAN ROWS; C/G: NATIVE/SOURCE COUNT; e_N, e_G: ABSOLUTE COUNT ERROR DIVIDED BY N, G; d_1, d_2: NATIVE ROW DISAGREEMENT WITH S1/S2. COMPLETE CALIBRATION-PLUS-EXECUTION COSTS ARE DIVIDED BY S1. A MISSING STRICT COUNT DOES NOT MEAN NATIVE FAILURE.

Workload	k	Native delivery	V_s/N	C/G	$e_N \mid e_G$ (%)	$d_1 \mid d_2$ (%)	Calls/S1	Raw tokens /S1
Qwen								
Movie top	1	yes	2048/2048	896/1,251	17.33 | 28.38	0.15 | 0.20	1.062	1.11
Movie other	1	yes	2048/2048	986/1,418	21.09 | 30.47	0.15 | 0.15	1.062	1.11
AG World	1	yes	114/2048	1,999/514	72.51 | 288.91	0.00 | 0.00	1.062	1.12
AG Sports	1	yes	387/2048	1,750/517	60.21 | 238.49	1.12 | 0.98	1.062	1.12
AG Business	1	yes	2040/2048	613/498	5.62 | 23.09	0.10 | 0.15	1.078	1.18
AG Sci/Tech	1	yes	2025/2048	483/519	1.76 | 6.94	0.34 | 0.24	1.062	1.12
GLM								
Movie top	1	yes	2048/2048	1,025/1,251	11.04 | 18.07	0.24 | 0.24	1.062	1.12
Movie other	1	yes	2048/2048	1,145/1,418	13.33 | 19.25	0.39 | 0.39	1.062	1.12
AG World	1	yes	2045/2048	621/514	5.22 | 20.82	0.24 | 0.20	1.086	1.21
AG Sports	8	yes	2048/2048	519/517	0.10 | 0.39	0.83 | 0.83	0.203	1.03
AG Business	1	yes	2045/2048	845/498	16.94 | 69.68	0.10 | 0.10	1.102	1.25
AG Sci/Tech	1	yes	2048/2048	499/519	0.98 | 3.85	0.15 | 0.05	1.062	1.12

TABLE IV

THREE POLICIES OVER THE SAME 12 PRIMARY CONFIGURATIONS. BENEFIT REQUIRES NATIVE DELIVERY, $k > 1$, BOTH REFERENCE LOSSES $\leq .05$, FEWER REQUESTS AND FEWER RAW INPUT-PLUS-OUTPUT TOKENS THAN S1. RANGES USE EACH CASE'S S1. STRICT BENEFIT RETAINS THE 12-CASE DENOMINATOR; IT IS UNAVAILABLE IF A FULL STRICT PLAN OR S1/S2 REFERENCE IS MISSING. FOR THE FIXED PLANS, ALL SIX UNAVAILABLE CASES ARISE FROM INCOMPLETE STRICT REFERENCES. FIXED PLANS HAVE NO CERTIFICATE; SOURCE CORRECTNESS IS SEPARATE.

Plan	Native delivery	Native $d_{\max}$ (%)	Calls/S1	Raw tokens/S1	Native benefit	Strict benefit; unavailable of 12
Random8	12/12	0.83–55.71	0.125–0.125	0.640–0.761	1/12	1/12; 6
Similar8	12/12	0.63–59.47	0.125–0.125	0.640–0.761	2/12	1/12; 6
Current80-S	12/12	0.00–1.12	0.203–1.102	1.030–1.251	0/12	0/12; 6

The other 35 main configurations select fresh singleton fallback and pay for calibration as well as their new complete execution. No secondary size/cache configuration adds a non-singleton selection. Sports was not a preselected scale anchor, so these measurements do not identify its break-even size outside $N = 2,048$.

D. More Probes, the Same Fallback Decisions

The follow-up holds the native-Boolean agreement rule fixed while comparing Current80-N with Current320-N. All 16 policies select singleton fallback and deliver native COUNT. Twelve have complete strict vectors; four have unavailable strict counts. Table V gives every paired comparison, and Fig. 3 separates checking from execution.

In all eight pairs, 320 probes increase full requests by 4.35–7.64%, raw tokens by 6.82–19.20%, and API uncached input by 8.69–25.46%. Neither tested budget improves the deployment choice on these anchors. The hypothetical IID power calculation in Section III does not estimate this panel's mismatch probability. Rejecting a candidate means that the policy did not certify it, not that every possible batch is unsafe.

One recorded pair makes the stopping bill concrete: Qwen AG Sci/Tech, $N = 7,600$, cache ON, repeat 1. In candidate order $8, 4, 2$, the 80-probe policy issues/scores 16, 16 and 48 batches, with 15, 15 and 47 agreements. Its first decisive disagreement is at probe 14, 13 and 33, leaving billed chunk tails of 2, 3 and 15 requests. At 320 probes the corresponding issued/scored counts are 48, 112 and 128, with 37, 104 and 120 agreements; the eighth disagreement occurs at 39, 101 and 119, leaving tails of 9, 11 and 9. All issued trials enter the recorded counts. Calibration is therefore $80 + 80 = 160$ versus $320 + 288 = 608$ requests. Both deploy 7,600 fresh singletons, yielding 7,760 versus 8,208 total requests. Table VI gives the absolute token receipts; phase-level receipts remain in the artifact.

Table VII records the stop evidence. The main study observes 106 candidate prefixes (105 rejections, one pass); the follow-up observes 48 rejections. At every rejected candidate prefix, accumulated native-Boolean mismatches alone make passing impossible. Qwen also has 31 probe-level strict-only format flags, classified after excluding batch failures and native mismatches. These flags do not mean that format alone caused a candidate stop. No issued probe has a batch-contract failure. The two lower candidates after the Sports pass remain unobserved. Stopped prefixes estimate neither population mismatch probability nor a false-rejection rate.

E. Cache Counters Reverse the Input-Cost Ranking

All 48 cache-ON fixed-plan executions (24 configurations, two plans) use less raw input but more API uncached input

TABLE V

EVERY PAIRED CURRENT80-N→CURRENT320-N FOLLOW-UP ($N = 7,600$, CACHE ON). R: COMPLETE REQUESTS; $d_{\max}$: MAXIMUM NATIVE ROW DISAGREEMENT WITH S1/S2. RAW-TOKEN AND UNCACHED-INPUT RATIOS COMPARE COMPLETE 320-PROBE WITH 80-PROBE POLICIES. REPEATS ARE DESCRIPTIVE; EACH BUDGET HAS ITS OWN NOMINAL SCOPE.

Model / workload	Rep.	k	Calibration calls	Full R	Raw tokens 320/80	Uncached input 320/80	$d_{\max}$ (%)
Qwen / AG Sci/Tech	1	1→1	160→608	7,760→8,208	1.11	1.13	0.41→0.36
Qwen / AG Sci/Tech	2	1→1	144→656	7,744→8,256	1.11	1.12	0.45→0.28
Qwen / Movie other	1	1→1	128→464	7,728→8,064	1.07	1.09	0.17→0.17
Qwen / Movie other	2	1→1	144→640	7,744→8,240	1.12	1.17	0.14→0.20
GLM / AG Sci/Tech	1	1→1	176→672	7,776→8,272	1.16	1.21	0.16→0.05
GLM / AG Sci/Tech	2	1→1	144→736	7,744→8,336	1.19	1.25	0.09→0.08
GLM / Movie other	1	1→1	128→464	7,728→8,064	1.07	1.10	0.20→0.20
GLM / Movie other	2	1→1	176→672	7,776→8,272	1.15	1.23	0.20→0.14

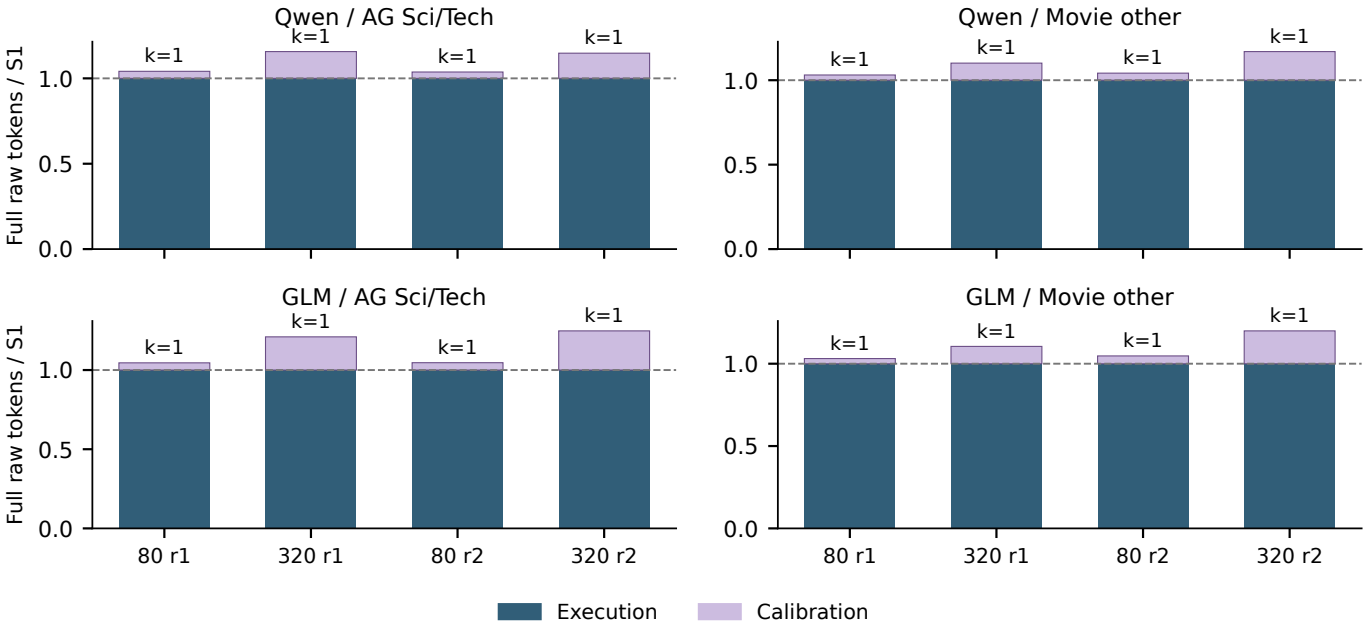


Fig. 3. All 16 native-rule follow-up policies ($N = 7,600$, cache ON). Each pair compares 80 and 320 probes within one descriptive physical repeat. Bars include fresh calibration and complete execution, divided by the original S1 input-plus-output token bill for that input. Every selected size is one. These measured costs are distinct from the hypothetical IID acceptance-power calculation.

TABLE VI

COMPLETE TOKEN RECEIPTS FOR THE WORKED QWEN AG SCI/TECH PAIR ($N = 7,600$, CACHE ON, REPEAT 1). P: RAW INPUT; O: OUTPUT; K: CACHED INPUT; $U = P - K$: UNCACHED INPUT. BOTH POLICIES DEPLOY FRESH SINGLETON FALLBACK. THESE ABSOLUTE COUNTERS ACCOMPANY THE PAIR'S REQUEST TRACE IN THE TEXT.

Budget	P	O	K	U
80	1,336,641	32,108	498,816	837,825
320	1,483,830	36,842	533,440	950,390

than their own S1. Qwen's ratios are .626–.739 for raw input versus 1.083–1.157 for uncached input; GLM's are .649–.758 versus 1.142–1.285. All 24 cache-OFF fixed-plan executions favor batching on both counters. Table VIII keeps raw input, output and uncached input separate, while Fig. 4 shows matched examples.

All 24 matched anchor/size/model/fixed-plan comparisons have identical raw input counts with cache ON and OFF. Since $U = P - K$, the ON singleton's cached-token advantage exceeds its extra raw input: $K_{S1} - K_{\text{batch}} > P_{S1} - P_{\text{batch}}$. This explains how lower batch input can coexist with higher uncached input. A raw-input-only cost model cannot reproduce that ranking. Output ratios remain separate: .974–1.106 for Qwen and approximately .850 for GLM with cache ON. These are counter rankings for the declared representation, not universal time or monetary advantages.

F. Classical Controls Can Accept COUNT Without Rows

Under the main strict-vector eligibility rule, the source-label raw-COUNT gate accepts 24/98 complete plan/configuration pairs; both the joint-strata and direct HG row gates accept

TABLE VII

OBSERVED CANDIDATE PREFIXES: O/P/R = OBSERVED/PASSED/REJECTED. FLAGS ARE CLASSIFIED B (INVALID FULL BATCH RESPONSE), THEN M (NATIVE-LABEL MISMATCH), THEN F (ADDITIONAL STRICT-ONLY REFERENCE-FORMAT REJECTION). AN M MAY ALSO HAVE A NON-STRICT REFERENCE. THE LAST COLUMN COUNTS STOPS ALREADY FORCED BY M ALONE AT THE FIXED FULL BUDGET. PROBES INCLUDE WHOLE ISSUED CHUNKS; TWO LOWER CANDIDATES AFTER THE SPORTS PASS ARE UNOBSERVED. STOPPED PREFIXES DO NOT ESTIMATE POPULATION MISMATCH PROBABILITY.

Family	Model	O/P/R	Issued probes	M flags	F flags	B flags	Native-sufficient stops / R
Current80-S	Qwen	54/0/54	1,040	143	31	0	54/54
Current80-S	GLM	52/1/51	1,264	101	0	0	51/51
Current80-N	Qwen	12/0/12	256	26	0	0	12/12
Current80-N	GLM	12/0/12	304	32	0	0	12/12
Current320-N	Qwen	12/0/12	1,088	108	0	0	12/12
Current320-N	GLM	12/0/12	1,264	104	0	0	12/12

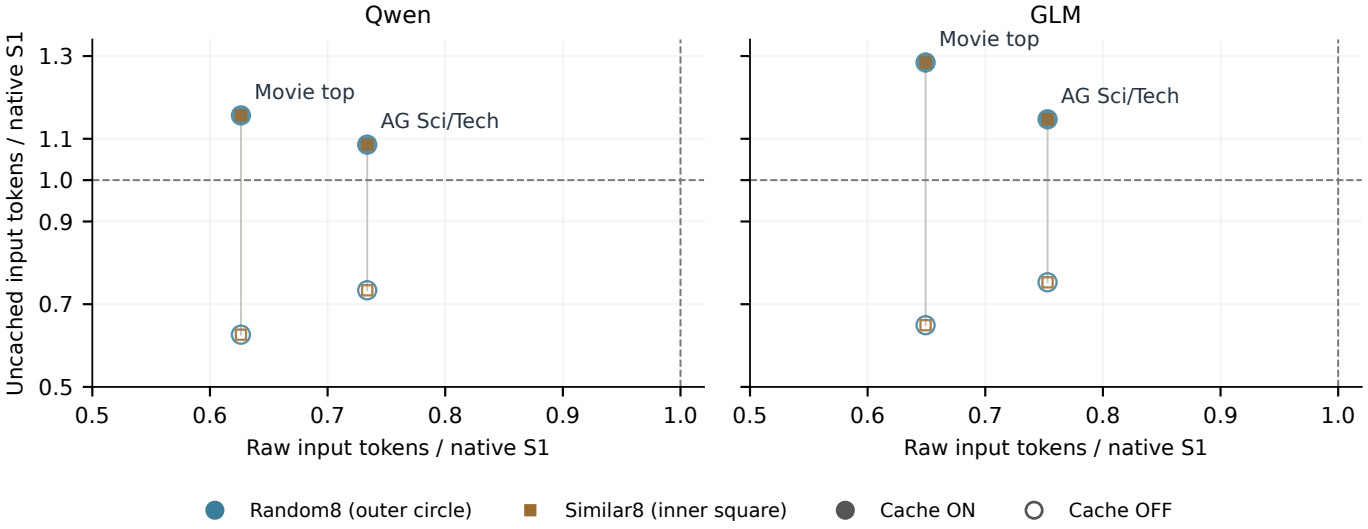


Fig. 4. Raw versus API uncached input for Movie top and AG Sci/Tech at $N = 7,600$. Lines connect matched cache-OFF/ON executions under each model. Ratios use the corresponding singleton S1 counter; one marks equal cost. Outer Random8 circles and inner Similar8 squares show coincident measured counters without jitter. Complete-matrix ranges are in Table VIII. Uncached input is a prefill-work proxy.

3/98 (Table IX). The 21 additional COUNT passes concern aggregate accuracy, not reusable row labels. No accepted source COUNT falls outside additive $.05N$ tolerance in these observed vectors. The post-hoc native-vector sensitivity evaluates 108 pairs, accepting 27 COUNTs and three row-gate passes. Its three extra COUNT passes are Qwen fallback vectors; it changes neither calibration nor deployment.

Direct AQP has a readout for every configuration regardless of batch completion. The same source-only estimate is reused across model/cache configurations sharing a workload, size and sample. Table X therefore shows 12 distinct workload/size pairs while the artifact retains all 36 configuration entries. Every interval has width at most $.1N$, but realized coverage is 32/36 entries, or 11/12 distinct pairs. Movie other at $N = 2,048$ is uncovered: $G = 1,418$, sample positives 732/1,024, HG interval $[1,423, 1,504]$. Its midpoint 1,463.5 gives $e_N = 2.22\%$ and $e_G = 3.21\%$. That sample is reused in four configurations. Width sufficiency and the repeated miss give no matrix-wide guarantee. The offline label lookups are not priced as native calls.

G. Reproducibility and the Research Ledger

Independent reconstruction performs 13,219,995 main-study and 4,059,851 follow-up checks with zero integrity errors or warnings. These concern execution-chain and resource reconstruction, not model correctness or the certificate's assumptions. Replaying the native agreement rule recovers the same selections and issued-call costs in all 36 main cases. Source hashes, request chains, cache-reset receipts, exact CSVs, plotting sources and native/strict readouts support CPU reconstruction without new judgments.

The successful-run ledger contains 557,996 calls: 429,584 main-study, 127,712 follow-up, 160 independent-validation, 480 original-Q3 and 60 format-control calls. It records 98,862,260 input tokens and 3,576,771 output tokens. The earlier supporting studies are outside this new-study ledger. The 160 independent-validation calls are outside Current80-S's standalone bill. Three preserved failed starts returned zero, zero and ten requests; their initialization work is unmetered. The ten returns add 814 input and 62 output tokens, giving 558,006 returned requests across successful and failed attempt-

TABLE VIII

FIXED-PLAN COST RATIOS TO THE CORRESPONDING SEPARATELY EXECUTED S1. CELLS GIVE COMPLETE-MATRIX RANGES; ON HAS 12 AND OFF SIX CONFIGURATIONS PER MODEL/PLAN. THE LAST COLUMN COUNTS RAW-INPUT RATIO $P/P_{S1} < 1$ BUT UNCACHED-INPUT RATIO $U/U_{S1} > 1$. INPUT, OUTPUT AND UNCACHED INPUT ARE DISTINCT; UNCACHED INPUT IS A PREFILL-WORK PROXY.

Model/cache	Plan	Cases	Raw input/S1	Output/S1	Uncached input/S1	$P/P_{S1} < 1$, $U/U_{S1} > 1$
Qwen/ON	Random8	12	0.63–0.74	0.98–1.11	1.08–1.16	12/12
Qwen/ON	Similar8	12	0.63–0.74	0.97–1.09	1.08–1.16	12/12
Qwen/OFF	Random8	6	0.63–0.73	1.06–1.08	0.63–0.73	0/6
Qwen/OFF	Similar8	6	0.63–0.73	1.06–1.08	0.63–0.73	0/6
GLM/ON	Random8	12	0.65–0.76	0.85–0.85	1.14–1.28	12/12
GLM/ON	Similar8	12	0.65–0.76	0.85–0.85	1.14–1.28	12/12
GLM/OFF	Random8	6	0.65–0.75	0.85–0.85	0.65–0.75	0/6
GLM/OFF	Similar8	6	0.65–0.75	0.85–0.85	0.65–0.75	0/6

TABLE IX

SOURCE-LABEL CONTROLS ACROSS ALL 36 CONFIGURATIONS (18 PER MODEL/PLAN). E: ELIGIBLE PLAN VECTORS; C/R/D: ACCEPTED RAW-COUNT/JOINT-STRATA-ROW/DIRECT-HG-ROW GATES. MAIN ELIGIBILITY REQUIRES COMPLETE STRICT PLAN LABELS, NOT STRICT SINGLETON REFERENCES. NATIVE COLUMNS ARE POST-HOC SENSITIVITY WITH THE SAME SAMPLE AND FIXED LOOK. EACH METHOD/PAIR HAS NOMINAL 95% SCOPE AT ADDITIVE $.05N$; COLUMNS ARE NOT SIMULTANEOUS CERTIFICATES.

Model	Plan	Canonical strict $E/18$	Strict C/R/D	Native sensitivity $E/18$	Native C/R/D
Qwen	Random8	18/18	0/0/0	18/18	0/0/0
Qwen	Similar8	18/18	0/0/0	18/18	0/0/0
Qwen	Current80-S	10/18	0/0/0	18/18	3/0/0
GLM	Random8	18/18	10/1/1	18/18	10/1/1
GLM	Similar8	18/18	8/1/1	18/18	8/1/1
GLM	Current80-S	16/18	6/1/1	18/18	6/1/1

TABLE X

DIRECT SOURCE-LABEL AQP: ALL 12 WORKLOAD/SIZE PAIRS, COLLAPSING REPEATED MODEL/CACHE ROLES FROM 36 CONFIGURATIONS. EACH SAMPLE GIVES A SEPARATE NOMINAL 95% HG INTERVAL. MIDPOINT ERRORS DIVIDE BY N AND G; REALIZED COVERAGE IS DISTINCT FROM INTERVAL WIDTH. COST IS m SOURCE-LABEL ORACLE QUERIES WITH NO FULL BATCH INFERENCE.

Workload	N	G	m	HG interval	Midpoint $e_N \mid e_G$ (%)	Covers G
Movie top	512	309	512	[309, 309]	0.00 | 0.00	yes
Movie top	2,048	1,251	1,024	[1,210, 1,297]	0.12 | 0.20	yes
Movie top	7,600	4,778	1,024	[4,685, 5,106]	1.55 | 2.46	yes
Movie other	512	359	512	[359, 359]	0.00 | 0.00	yes
Movie other	2,048	1,418	1,024	[1,423, 1,504]	2.22 | 3.21	no
Movie other	7,600	5,197	1,024	[5,109, 5,512]	1.49 | 2.18	yes
AG World	2,048	514	1,024	[466, 544]	0.44 | 1.75	yes
AG Sports	2,048	517	1,024	[457, 533]	1.07 | 4.26	yes
AG Business	2,048	498	1,024	[466, 544]	0.34 | 1.41	yes
AG Sci/Tech	512	123	512	[123, 123]	0.00 | 0.00	yes
AG Sci/Tech	2,048	519	1,024	[484, 562]	0.20 | 0.77	yes
AG Sci/Tech	7,600	1,900	1,024	[1,779, 2,164]	0.94 | 3.76	yes

s. This research ledger includes evaluation work; the policy tables charge each policy's own calibration and execution.

Materials and run instructions are available in the public repository, including source/acquisition records, execution receipts, exported readouts, plotting code and reconstruction/build instructions.

VII. EARLIER SUPPORTING EVIDENCE

A separate earlier replication holds a target fixed while changing its neighboring fillers, rather than executing a complete query partition. It implements established interference controls [4] on 128 targets each for DBLP–ACM, BoolQ and SNLI, using Qwen, SmolLM2-1.7B and GLM [12], [13], [14], [15], [9], [10], [11]. The summarized prompt condition requests the target label before filler labels. For each target, the analysis subtracts replay disagreement within a fixed filler condition from disagreement after replacing near fillers with random fillers. Five of nine task/model cases have a strictly positive lower endpoint in the retained target-level bootstrap interval for this excess; four do not. Failure to pass is not

equivalence, and these approximate intervals are not exact population or family-wise guarantees. The artifact preserves the complete table, precise repeat pairing and bootstrap protocol, output-contract controls and every negative case. These probes do not isolate the mechanism of the present integrated plan comparison.

The retained development-transfer study tests reuse of a policy chosen on development data, rather than current-input certification. Its 23 relations comprise three DBLP and 20 BoolQ relations. Most .05/.10 reference-row settings do not recover calibration costs within this panel. In the Qwen BoolQ subset at the secondary .20 tolerance, one policy uses 1,152 development-calibration calls plus 640 deployment calls = 1,792, versus 2,560 singleton calls, but about 1.69 times its tokens. Within these 20 BoolQ relations, 17 meet that .20 reference-row tolerance in both replays. The complete selector grid and all negative costs remain in the artifact. This bounded cost example supplies neither a source-relative COUNT guarantee nor an additional result in the new 36-configuration matrix.

VIII. Related Work

BATCHER explores grouping and demonstrations for entity resolution [1]; BatchPrompt studies ordering, ensembles and early stopping [2]; multi-instance processing separates instance count from context length [3]. Company distinguishes neighboring-text and prior-verdict channels, shows aggregate masking of row changes and discusses cached singleton prefixes [4]. Hassanzadeh and Subramanian measure row/COUNT changes, repetition and cascade calibration in semantic queries [8]. These precedents motivate the present controls; our live plan comparison additionally changes serialization and output contracts.

Guarantees also differ in their preservation targets. LOTUS optimizes relative to reference algorithms [5]; SUPG targets precision/recall using proxies and oracle samples [18]. BARGAIN develops guaranteed LLM cascades, including adaptive sampling without replacement and valid sample reuse [22]. NousDB supplies the independent-probe current-input CP procedure reproduced here [6]. Its Table 3 already reports Certified/Batched token/runtime costs, Fig. 11 reports calibration and small selected sizes, and §5.2 discusses continuous batching and prefix caching for efficient per-row execution. Our contribution is the joined execution audit of that paper-defined subprocedure under the declared maximum size, agreement rule and deployment representation.

The COUNT controls follow established AQP. SampleClean's NormalizedSC corrects a full dirty-data result using a cleaned sample and also supports direct clean-sample estimation [17]. BlazeIt uses aggregation control variates [19]; ABAE uses proxy strata and sample allocation for expensive predicates [20]. ThalamusDB combines model predicates, partial evaluation and query rewrites for deterministic approximate-query bounds [21], with a different complete-model target and bound construction. Finite-population inference and confidence sequences address fixed populations sampled without replacement [23]; the present controls use one fixed look.

SemBench measures real engines' source-defined COUNT quality, cost, latency and scale [7]. We retain original Q3 as an integration case and identify the larger inputs as source-derived variants. The empirical focus is how fidelity checking, a complete deployed plan and its complete bill interact when cache state and probe budget are measured explicitly.

IX. Implications and Scope

The execution record supports two optimizer decisions: name the preservation target, then compare the complete bill under the actual cache state. Native-reference fidelity, reusable row labels and source COUNT call for different checks. COUNT-only applications can compare aggregate controls and direct AQP under their oracle requirements. A cost catalog should retain requests, input, output, cache receipts and all checking phases: calibration changes the Sports raw-token decision, while cache state changes the fixed plans' input-cost ordering.

The measured scope is two local models, six fixed predicates, two domains and three prepared views. Shared AG records, nested sizes and descriptive budget repeats retain their dependencies. The Qwen panels characterize the frozen wording/model/parser combination jointly. Batching repeats each complete context/claim and predicate while changing JSON serialization, system instructions and output requirements. Its cache ordering is therefore specific to this representation; factored templates and isolated context-interference effects require separate comparisons. The accepted deployment meets observed tolerance, leaving certificate transport failure unobserved.

Fixed-vector intervals refer to supplied annotations and their named immutable vectors. Each nominal 95% procedure has its stated per-pair scope, separate from winner selection or future replays. Uncached input measures a prefill-work proxy; shared-server timing includes tracing and isolated clients. These resource objects support the reported plan comparisons, with monetary cost, general latency and a full NousDB optimizer comparison outside the measured scope.

X. Conclusion

The study connects fidelity checking to complete query results and resource bills. Fixed plans meet the joint reference-and-cost criterion in a few primary configurations; the checked policy meets it in none, including its fidelity-preserving Sports deployment. Larger probe budgets retain fallback and raise costs on the follow-up. Native cache counters reverse raw-input ordering for the declared cache-enabled fixed plans. The execution record enables reproducible comparisons of preservation targets and complete policy costs.

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